# VLSI Circuits for DSP

**Course Structure**: 3 - 0 - 4 - 5
**Course Code:**
**Prerequisite Courses**:

**Course Objective**

Digital signal processing hardware is an important element in real time application that enhances the performance of the system. This course provides various VLSI architectures used in digital signal processing such as discrete orthogonal transformations, digital filtering etc. The course aims to provide advanced understanding of hardware designs of various Digital Signal Processing operations.

**Course Outcome:** At the end of the course, students will be able to:

- Understand the hardware and design concepts of DSP filtering techniques and transformations and their circuit implementations

**Syllabus**

- Arithmetic Circuits
- Building blocks for DSP Processors
- Digital filter design
- DFT architectures
- Design of DSP Data - Path Blocks

**Textbooks:**

- V. Oppenheium - R.W. Schafer, Discrete-Time Sig Processing ( 2nd Ed, Prentice Hall, 1999)
- M. Mano - Computer System Architecture ( 3rd Edition, Pearson Publication, 2007 )
- John L. Hannessey & David A. Patterson - Computer Architecture: A Quantitative Approach ( 4th Edition, Elsevier, Morgan Kaufmann Publishers, 2007 )
- DSP, A Practical Approach by Ifeachor and Jervis
- Y.T. Chan - Wavelet Basics (Kluwer Publishers, Boston, 1993)